Height-preserving invertibility of the G3c propositional rules

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Editorial note. Scope declaration: the author proves the left-and part in full and frames the opening sentence as for any rule; the remaining seven parts are uniform instances of the same three-branch argument (axiom atomicity, principal immediacy, non-principal inductive hypothesis plus rule re-application), which this card declares rather than re-derives.

Editorial note. The exercise printed after the proof asking for the right-implies part lies outside the counted span.

Editorial scope. One original lemma of the unit's inversion section, counted once: the eight-part height-preserving inversion of the G3c propositional rules. The counted lemma and its author proof are retained.

Allowed standard prerequisites (editorial).

- The one-sided classical sequent calculus G3c: atomic axioms of the form C on the left and C on the right with everything else in the contexts, the propositional rules for not, and, or, and implies, and the notion of the height of a proof.
- The induction-with-varying-contexts convention: in height-preserving arguments the inductive hypothesis is stated for all contexts, not just the ones occurring in the given proof.

Lemma 0.1. *The rules of G3c for $\neg$, $\wedge$, $\vee$, and $\rightarrow$ are height-preserving invertible, that is:*

1. *: If* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, \neg A$*, then* $\mathbf{G3c} \vdash [n]A, \Gamma \Rightarrow \Delta$*.*
2. *: If* $\mathbf{G3c} \vdash [n]\Gamma, \neg A \Rightarrow \Delta$*, then* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A$*.*
3. *: If* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A \wedge B$*, then* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A$ *and* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, B$*.*
4. *: If* $\mathbf{G3c} \vdash [n]\Gamma, A \wedge B \Rightarrow \Delta$*, then* $\mathbf{G3c} \vdash [n]A, B, \Gamma \Rightarrow \Delta$*.*
5. *: If* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A \vee B$*, then* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A, B$*.*
6. *: If* $\mathbf{G3c} \vdash [n]A \vee B, \Gamma \Rightarrow \Delta$*, then* $\mathbf{G3c} \vdash [n]A, \Gamma \Rightarrow \Delta$ *and* $\mathbf{G3c} \vdash [n]B, \Gamma \Rightarrow \Delta$*.*

7. : *If* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A \to B$, *then* $\mathbf{G3c} \vdash [n]A, \Gamma \Rightarrow \Delta, B$.

8. : *If* $\mathbf{G3c} \vdash [n]A \to B, \Gamma \Rightarrow \Delta$, *then* $\mathbf{G3c} \vdash [n]\Gamma \Rightarrow \Delta, A$ *and* $\mathbf{G3c} \vdash [n]B, \Gamma \Rightarrow \Delta$.

Proof. We have to show that, for any rule R of G3c, if there is proof π of the conclusion S of R, then there are also proofs π_i of the premises S_i of R, with $\operatorname{ht}\pi_i \leq \operatorname{ht}\pi$. Consider the rule : suppose there is proof π with end-sequent of the form $A \wedge B, \Gamma \Rightarrow \Delta$. We have to show that there is proof π' of $A, B, \Gamma \Rightarrow \Delta$ with $\operatorname{ht}\pi' \leq \operatorname{ht}\pi$. We do this by induction on $n = \operatorname{ht}\pi$.

If $n = 0$, π consists of just an axiom: $A \wedge B, \Gamma \Rightarrow \Delta$. In G3c, axioms are of the form $C, \Gamma \Rightarrow \Delta, C$ with C atomic. In other words, C can't be $A \wedge B$. If $A \wedge B, \Gamma \Rightarrow \Delta$ is an axiom, then some atomic C must be element of both Γ and Δ. But then $A, B, \Gamma \Rightarrow \Delta$ is also an axiom, and we have found our proof π' (which is also of height 0).

If $n > 0$ it ends in an inference. We have to prove the claim for n, but may assume that it is true for any proof of height $< n$, even if the context is different. In other words, for any $k < n$ and any Π and Λ, if $A \wedge B, \Pi \Rightarrow \Lambda$ has proof of height k, then there is proof of height $\leq k$ of $A, B, \Pi \Rightarrow \Lambda$.

We have two cases: either $A \wedge B$ on the left is principal, or it is not. If it is principal, the last inference is and the immediate sub-proof π_1 ends in $A, B, \Gamma \Rightarrow \Delta$. The result holds in this case as $\operatorname{ht}\pi_1 < \operatorname{ht}\pi$.

If $A \wedge B$ is not principal, some other formula is and $A \wedge B$ belongs to the context of the last inference. Then the inductive hypothesis applies to the immediate sub-proofs of the last rule R, which yields proof π_1' or two proofs π_1' and π_2' of height no greater than the height(s) of the immediate sub-proofs of π. The end-sequents of π_1' and π_2' contain A, B in the context on the left instead of $A \wedge B$. Apply the rule R to obtain the proof π'. For instance, suppose π ends in :

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1 & \vdots\,\pi_2 \\ A \wedge B, \Gamma'\Delta, C & A \wedge B, \Gamma', D\Delta\end{array}}{A \wedge B, \Gamma', C \to D\Delta}$$

Here $C \to D$ on the left is principal, and the context on the left is $A \wedge B, \Gamma'$ (i.e., $\Gamma = \Gamma', C \to D$). The inductive hypothesis yields proofs π_1' and π_2' of $A \wedge B, \Gamma' \Rightarrow \Delta, C$ and $A \wedge B, \Gamma', D \Rightarrow \Delta$, respectively. These two sequents can again serve as premises for the rule to get π':

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1' & \vdots\,\pi_2' \\ A, B, \Gamma'\Delta, C & A, B, \Gamma', D\Delta\end{array}}{A, B, \Gamma', C \to D\Delta}$$

We have

$$\operatorname{ht}\pi = \max(\operatorname{ht}\pi_1, \operatorname{ht}\pi_2) + 1$$
$$\operatorname{ht}\pi' = \max(\operatorname{ht}\pi_1', \operatorname{ht}\pi_2') + 1$$

by definition, and

$$\begin{aligned} \operatorname{ht} \pi_1' &\leq \operatorname{ht} \pi_1 \\ \operatorname{ht} \pi_2' &\leq \operatorname{ht} \pi_2 \end{aligned}$$

by inductive hypothesis. So,

$$\operatorname{ht} \pi' \leq \operatorname{ht} \pi.$$

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